The search results provide useful information on rubber composite formulations for hydro-seals and some raw material prices, but a complete, itemized price list for all compounding agents is not available. Below is a synthesis of what can be confirmed from the search results.

□ Rubber Composite Materials for Hydro-Power Hydro-Seals

Modern hydro-seal materials have evolved beyond simple natural rubber. Based on recent patents and technical reports, the following composite systems are used:

| Composite System | Key Components | Performance Features |

|:---|:---|:---|

| **EPDM/Brominated Butyl Rubber Blend** | EPDM, brominated butyl rubber, activators, anti-aging agents, reinforcing fillers, plasticizers, tackifiers, co-vulcanizing agents | High interfacial adhesion, excellent weather aging resistance, good low-temperature toughness; solves traditional EPDM's poor polarity and adhesion issues |

| **EPDM/Silicone/Fluororubber Ternary Blend** | EPDM (50-80 phr), phenyl silicone rubber (10-30 phr), fluororubber (10-30 phr), compatibilizer, reinforcing filler, thermal conductive filler, coupling agent | Wide temperature range, long service life, improved aging resistance under large temperature differentials |

| **Brominated Butyl Rubber (BIIR) Based** | BIIR as base, carbon black (50 phr), naphthenic oil (50 phr), tackifying resins (rosin, coumarone, phenolic) | Excellent air tightness and self-adhesive properties; rosin at 70 phr gives optimal peel strength of 41.77 N/25mm |

□ Various Mixing Agents (Compounding Ingredients)

The following compounding agents are required for hydro-seal rubber formulations:

Vulcanization System:

- **Zinc oxide** (1-5 phr): Curing activator

- **Stearic acid** (0.5-3 phr): Co-activator

- **Sulfur** (0-2 phr): Crosslinking agent

- **Peroxide (DCP)** (2-5 phr): Crosslinking agent for EPDM

- **Co-agent TAIC** (2-5 phr): Co-crosslinking agent

- **Composite vulcanization aids** (4-6.94 phr): Sulfur + peroxide + bridging agent + accelerator

Accelerators: DM, NS, CZ, TMTD, TT, DPTT (0-3 phr)

Reinforcing Fillers:

- **Carbon black** (30-80 phr): N330, N550, N660 grades

- **Silica** (white carbon black): Used in some formulations

Anti-Aging Agents: 4010NA, 4020, 3100 (3-5 phr)

Plasticizers: Paraffin oil, aromatic oil, naphthenic oil, liquid EPDM (10-30 phr)

Tackifiers: Rosin, coumarone resin, phenolic resin

Coupling Agents: Silane coupling agents (0.5-3 phr)

Special Additives: Brominated butyl rubber for adhesion improvement, methylene acceptor/donor systems

□ Prices in China (Available Data)

Raw Material Prices (2026):

Material	Price (RMB/ton)	Source
EPDM 4045	21,950 – 22,150	
EPDM J-4045 (Hengshui)	26,750	
Carbon black N330	6,500	
Natural rubber (STR20 blend)	15,535	

EPDM Import vs. Domestic:

- Imported EPDM: ~28,900 RMB/ton
- Domestic EPDM: ~18,000 RMB/ton
- Price gap: >60%

Finished Hydro-Seal Product Prices:

Product	Price	Source
P-type rubber waterstop (per unit)	¥40	
External corner waterstop (per meter)	¥20/meter	
P-type rubber sealing strip (per meter)	¥50/meter	
Gate rubber seal (per piece)	¥15	

□ Summary

For hydro-power hydro-seal packing, the trend is toward **EPDM-based blends** with brominated butyl rubber for adhesion, or EPDM/silicone/fluororubber ternary systems for extreme temperature ranges. The key compounding agents include vulcanization systems (ZnO, stearic acid, sulfur/peroxide, accelerators), reinforcing fillers (carbon black, silica), anti-aging agents, plasticizers, and tackifiers. Raw material prices in China for EPDM range from approximately **18,000–28,900 RMB/ton**, with significant premiums for imported grades. Finished hydro-seal products range from **¥15–50 per piece or meter** depending on type and specification.

Note: The search results do not provide prices for all individual compounding agents (e.g., specific anti-aging agents, accelerators, tackifiers). These are typically purchased in smaller quantities and prices fluctuate. For a complete cost analysis, contacting local rubber chemical suppliers in China is recommended.